WORKED SOLUTION

STEP I 1988, Question 2

Pure • Stationary Points

> **STEP I 1988, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/88-s1-q2

The key idea

Think of g as a composition: $\mathrm{g} = h \circ \mathrm{f}$ with $h(t) = t + 1/t$. The chain rule then says g' vanishes either where $\mathrm{f}' = 0$ or where $h'(\mathrm{f}) = 0$, that is, where $\mathrm{f} = \pm 1$. Writing $\cos 2x$ in terms of $\sin x$ turns f into a quadratic in $\sin x$, which tells us everything about its size and shape.

Solution

The derivatives

Differentiating $\mathrm{g} = \mathrm{f} + \mathrm{f}^{-1}$ with the chain rule,

$$\mathrm{g}'(x) = \mathrm{f}'(x) - \frac{\mathrm{f}'(x)}{\mathrm{f}(x)^2} = \mathrm{f}'(x)\left(1 - \frac{1}{\mathrm{f}(x)^2}\right).$$

Differentiating again with the product rule, and using $\dfrac{\mathrm{d}}{\mathrm{d}x}(-\mathrm{f}^{-2}) = 2\mathrm{f}^{-3}\mathrm{f}'$,

$$\mathrm{g}''(x) = \mathrm{f}''(x)\left(1 - \frac{1}{\mathrm{f}(x)^2}\right) + \frac{2\,\mathrm{f}'(x)^2}{\mathrm{f}(x)^3}.$$

The size of f

Using $\cos 2x = 1 - 2\sin^2 x$ and writing $s = \sin x$,

$$\mathrm{f}(x) = 5 + 2s - 2s^2 = \tfrac{11}{2} - 2\left(s - \tfrac{1}{2}\right)^2.$$

As s runs over $[-1, 1]$, this is largest ($\frac{11}{2}$) at $s = \frac{1}{2}$ and smallest at the end of the interval furthest from $\frac{1}{2}$, namely $s = -1$, where it equals 1. So

$$1 \le \mathrm{f}(x) \le \tfrac{11}{2} \quad \text{for all } x,$$

with $\mathrm{f}(x) = 1$ only when $\sin x = -1$. In particular f is never zero, so g is defined everywhere.

Stationary points

$g'(x) = 0$ exactly when $f'(x) = 0$ or $f(x)^2 = 1$.

Zeros of f'. We have

$$f'(x) = -2\sin 2x + 2\cos x = 2\cos x\,(1 - 2\sin x),$$

which vanishes when $\cos x = 0$ or $\sin x = \frac{1}{2}$; in $[0, 2\pi]$ that is

$$x = \tfrac{\pi}{6},\ \tfrac{\pi}{2},\ \tfrac{5\pi}{6},\ \tfrac{3\pi}{2}.$$

Points where $f = \pm 1$. Since $f \geq 1$, this means $f = 1$, i.e. $\sin x = -1$, i.e. $x = \frac{3\pi}{2}$, which is already on the list.

So g has exactly four stationary points in $[0, 2\pi]$. (The endpoints are not among them: $g'(0) = g'(2\pi) = 2 \cdot \frac{24}{25} \neq 0$.)

Classifying them

We have $f''(x) = -4\cos 2x - 2\sin x$. At each point $f' = 0$, so the second term of g'' drops out and $g'' = f''\,(1 - f^{-2})$.

- $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$: here $\sin x = \frac{1}{2}$, $\cos 2x = \frac{1}{2}$, so $f = \frac{11}{2}$ and $f'' = -2 - 1 = -3$. Then $g'' = -3\left(1 - \frac{4}{121}\right) < 0$: both are **maxima**, with

$$g = \tfrac{11}{2} + \tfrac{2}{11} = \tfrac{125}{22}.$$

- $x = \frac{\pi}{2}$: here $\sin x = 1$, $\cos 2x = -1$, so $f = 5$ and $f'' = 4 - 2 = 2$. Then $g'' = 2\left(1 - \frac{1}{25}\right) = \frac{48}{25} > 0$: a **minimum**, with $g = 5 + \frac{1}{5} = \frac{26}{5}$.
- $x = \frac{3\pi}{2}$: here $f = 1$, so $1 - f^{-2} = 0$ and also $f' = 0$. Hence $g'' = 0$.

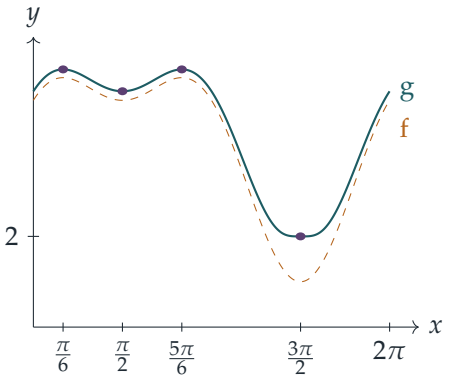
> **Ask yourself** The second-derivative test fails at $x = \frac{3\pi}{2}$. How else can we decide?

Go back to $h(t) = t + 1/t$. For $t > 0$,

$$t + \frac{1}{t} - 2 = \frac{(t-1)^2}{t} \geq 0,$$

with equality only at $t = 1$. Since $f > 0$, this gives $g(x) \geq 2$ for every x, with equality exactly when $f(x) = 1$. At $x = \frac{3\pi}{2}$ we have $g = 2$, so this is a **minimum** (indeed the global minimum of g).

Summary. Maxima at $x = \frac{\pi}{6}, \frac{5\pi}{6}$ (value $\frac{125}{22}$); minima at $x = \frac{\pi}{2}$ (value $\frac{26}{5}$) and $x = \frac{3\pi}{2}$ (value 2).



Checks

- Since h is increasing for $t > 1$, g should rise and fall with f wherever $\mathrm{f} > 1$; the graph confirms that f (dashed) and g have the same turning points.
- Numerically, $\mathrm{g}(\frac{\pi}{6}) \approx 5.6818 = \frac{125}{22}$ and $\mathrm{g}(\frac{\pi}{2}) = 5.2$; a fine scan of g over $[0, 2\pi]$ finds turning points only at $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$, with minimum value 2.000.
- The formula for g'' agrees with a finite-difference second derivative at random points.

Common slips

- Forgetting the factor $1 - \mathrm{f}^{-2}$: here it adds no new stationary point, but only because $\mathrm{f} = 1$ happens where $\mathrm{f}' = 0$ anyway. You must still check it.
- Concluding from $\mathrm{g}''(\frac{3\pi}{2}) = 0$ that the point is a point of inflection. A zero second derivative tells you nothing on its own.

Beyond STEP

The degenerate point $x = \frac{3\pi}{2}$ is typical of compositions: if f has a minimum where h also has a minimum, $h \circ \mathrm{f}$ is "flatter" there, and the second derivative vanishes. Undergraduate analysis replaces the second-derivative test with Taylor's theorem, which classifies such points by the first non-zero derivative, or (as we did) with a direct inequality. Abbott's *Understanding Analysis* is a good place to see the mean value theorem and Taylor's theorem developed carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.